STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-LQC-COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1977-2006 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2008-2106, the complete author proof (97 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_algebraic-stacks-sheaves.json (pb=4649, stmt 1977-2006, proof 2008-2106).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GNB: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2106–2106. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E8H: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2106–2106. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GN2: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2106–2106. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03EL: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2106–2106. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids.*

(1) *The category $LQCoh(\mathcal{O}_\mathcal{X})$ has colimits and they agree with colimits in the category $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$.*
(2) *The category $LQCoh(\mathcal{O}_\mathcal{X})$ is abelian with kernels and cokernels computed in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$, in other words the inclusion functor is exact.*

Date: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(3) *Given a short exact sequence* $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ *of* $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ *if two out of three are locally quasi-coherent so is the third.*
(4) *Given* $\mathcal{F}, \mathcal{G}$ *in* $LQCoh(\mathcal{O}_\mathcal{X})$ *the tensor product* $\mathcal{F} \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{G}$ *in* $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ *is an object of* $LQCoh(\mathcal{O}_\mathcal{X})$.
(5) *Given* $\mathcal{F}, \mathcal{G}$ *in* $LQCoh(\mathcal{O}_\mathcal{X})$ *with* $\mathcal{F}$ *of finite presentation on* $\mathcal{X}_{étale}$ *the sheaf* $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})$ *in* $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ *is an object of* $LQCoh(\mathcal{O}_\mathcal{X})$.

Proof. In the arguments below x denotes an arbitrary object of $\mathcal{X}$ lying over the scheme U. To show that an object $\mathcal{H}$ of $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ is in $LQCoh(\mathcal{O}_\mathcal{X})$ we will show that the restriction $x^*\mathcal{H}|_{U_{étale}} = \mathcal{H}|_{U_{étale}}$ is a quasi-coherent object of $Mod(U_{étale}, \mathcal{O}_U)$.

Proof of (1). Let $\mathcal{I} \to LQCoh(\mathcal{O}_\mathcal{X})$, $i \mapsto \mathcal{F}_i$ be a diagram. Consider the object $\mathcal{F} = \operatorname{colim}_i \mathcal{F}_i$ of $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$. The pullback functor x^* commutes with all colimits as it is a left adjoint. Hence $x^*\mathcal{F} = \operatorname{colim}_i x^*\mathcal{F}_i$. Similarly we have $x^*\mathcal{F}|_{U_{étale}} = \operatorname{colim}_i x^*\mathcal{F}_i|_{U_{étale}}$. Now by assumption each $x^*\mathcal{F}_i|_{U_{étale}}$ is quasi-coherent. Hence $\operatorname{colim}_i x^*\mathcal{F}_i|_{U_{étale}}$ is quasi-coherent by Descent, Lemma 0GNB. Thus $x^*\mathcal{F}|_{U_{étale}}$ is quasi-coherent as desired.

Proof of (2). It follows from (1) that cokernels exist in $LQCoh(\mathcal{O}_\mathcal{X})$ and agree with the cokernels computed in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of $LQCoh(\mathcal{O}_\mathcal{X})$ and let $\mathcal{K} = \operatorname{Ker}(\varphi)$ computed in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$. If we can show that $\mathcal{K}$ is a locally quasi-coherent module, then the proof of (2) is complete. To see this, note that kernels are computed in the category of presheaves (no sheafification necessary). Hence $\mathcal{K}|_{U_{étale}}$ is the kernel of the map $\mathcal{F}|_{U_{étale}} \to \mathcal{G}|_{U_{étale}}$, i.e., is the kernel of a map of quasi-coherent sheaves on $U_{étale}$ whence quasi-coherent by Descent, Lemma 0GNB. This proves (2).

Proof of (3). Let $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ be a short exact sequence of $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$. Since we are using the étale topology, the restriction $0 \to \mathcal{F}_1|_{U_{étale}} \to \mathcal{F}_2|_{U_{étale}} \to \mathcal{F}_3|_{U_{étale}} \to 0$ is a short exact sequence too. Hence (3) follows from the corresponding statement in Descent, Lemma 0GNB.

Proof of (4). Let $\mathcal{F}$ and $\mathcal{G}$ be in $LQCoh(\mathcal{O}_\mathcal{X})$. Since restriction to $U_{étale}$ is given by pullback along the morphism of ringed topoi $U_{étale} \to (Sch/U)_{étale} \to \mathcal{X}_{étale}$ we see that the restriction of the tensor product $\mathcal{F} \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{G}$ to $U_{étale}$ is equal to $\mathcal{F}|_{U_{étale}} \otimes_{\mathcal{O}_U} \mathcal{G}|_{U_{étale}}$, see Modules on Sites, Lemma 03EL. Since $\mathcal{F}|_{U_{étale}}$ and $\mathcal{G}|_{U_{étale}}$ are quasi-coherent, so is their tensor product, see Descent, Lemma 0GNB.

Proof of (5). Let $\mathcal{F}$ and $\mathcal{G}$ be in $LQCoh(\mathcal{O}_\mathcal{X})$ with $\mathcal{F}$ of finite presentation. Since $(Sch/U)_{étale} = \mathcal{X}_{étale}/x$ is a localization of $\mathcal{X}_{étale}$ at an object we see that the restriction of $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})$ to $(Sch/U)_{étale}$ is equal to

$$\mathcal{H} = \mathcal{H}om_{\mathcal{O}|_{(Sch/U)_{étale}}}(\mathcal{F}|_{(Sch/U)_{étale}}, \mathcal{G}|_{(Sch/U)_{étale}})$$

by Modules on Sites, Lemma 0E8H. The morphism of ringed topoi $(U_{étale}, \mathcal{O}_U) \to ((Sch/U)_{étale}, \mathcal{O})$ is flat as the pullback of $\mathcal{O}$ is $\mathcal{O}_U$. Hence the pullback of $\mathcal{H}$ by this morphism is equal to $\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_{U_{étale}}, \mathcal{G}|_{U_{étale}})$ by Modules on Sites, Lemma 0GN2. In other words, the restriction of $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})$ to $U_{étale}$ is $\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_{U_{étale}}, \mathcal{G}|_{U_{étale}})$. Since $\mathcal{F}|_{U_{étale}}$ and $\mathcal{G}|_{U_{étale}}$ are quasi-coherent, so is $\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_{U_{étale}}, \mathcal{G}|_{U_{étale}})$, see Descent, Lemma 0GNB. We conclude as before. □